



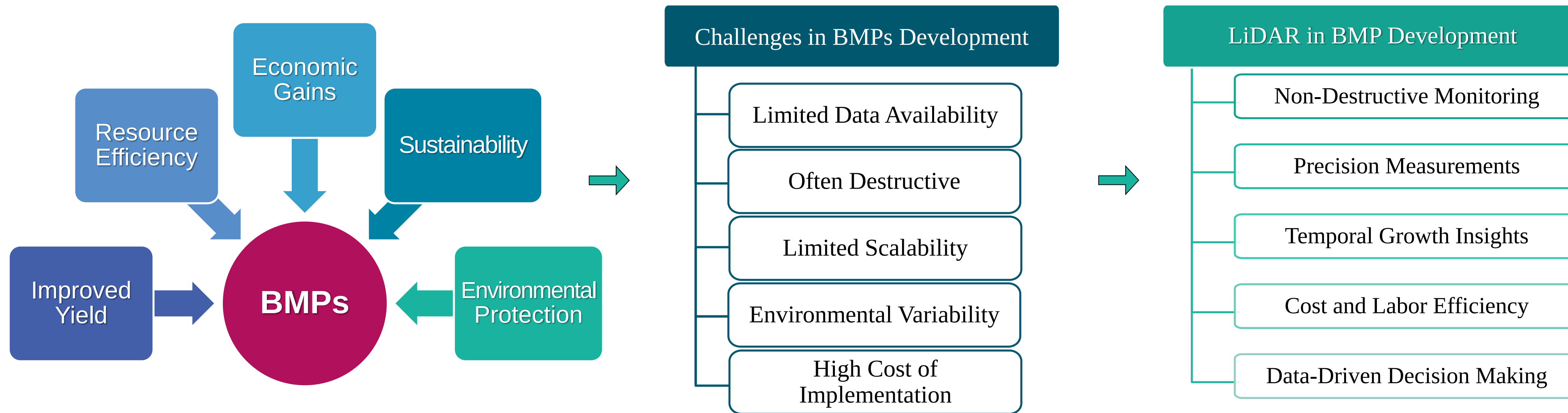
# Development of Remote Sensing Tools to Evaluate In-Field Results of Best Management Practices (BMPs) for Peas

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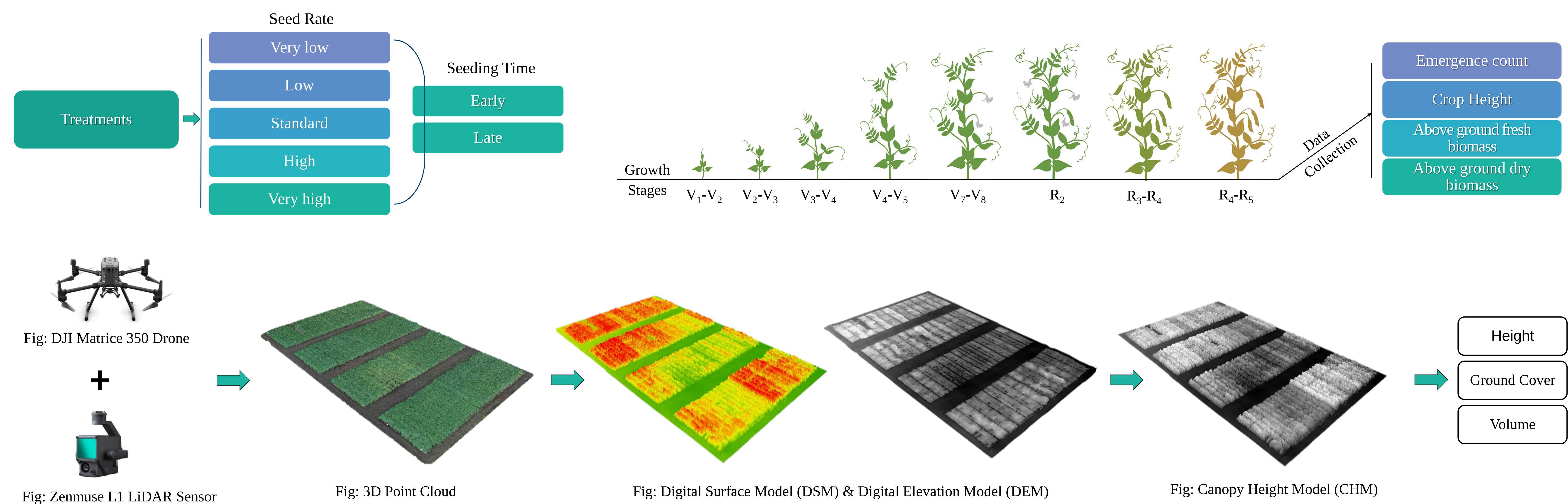
## Introduction



## Objectives

- Develop structural remote sensing indices to capture pea growth and development over time.
- Determine the optimal time for crop growth stages to best predict crop yield from remote sensing data.

## Methodology



## Result

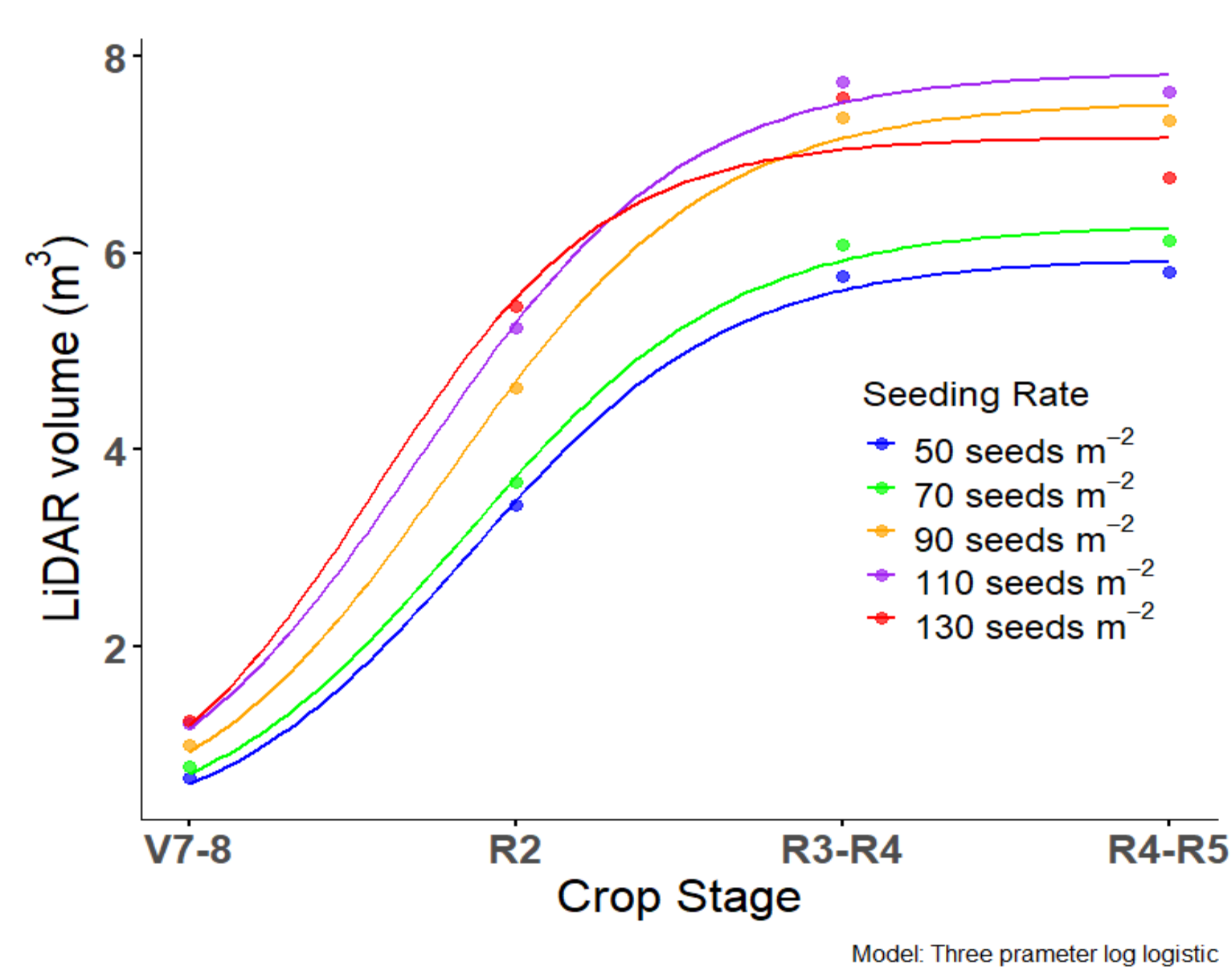


Figure 1: LiDAR Point Cloud Derived Volume Accumulation over time by seeding rate

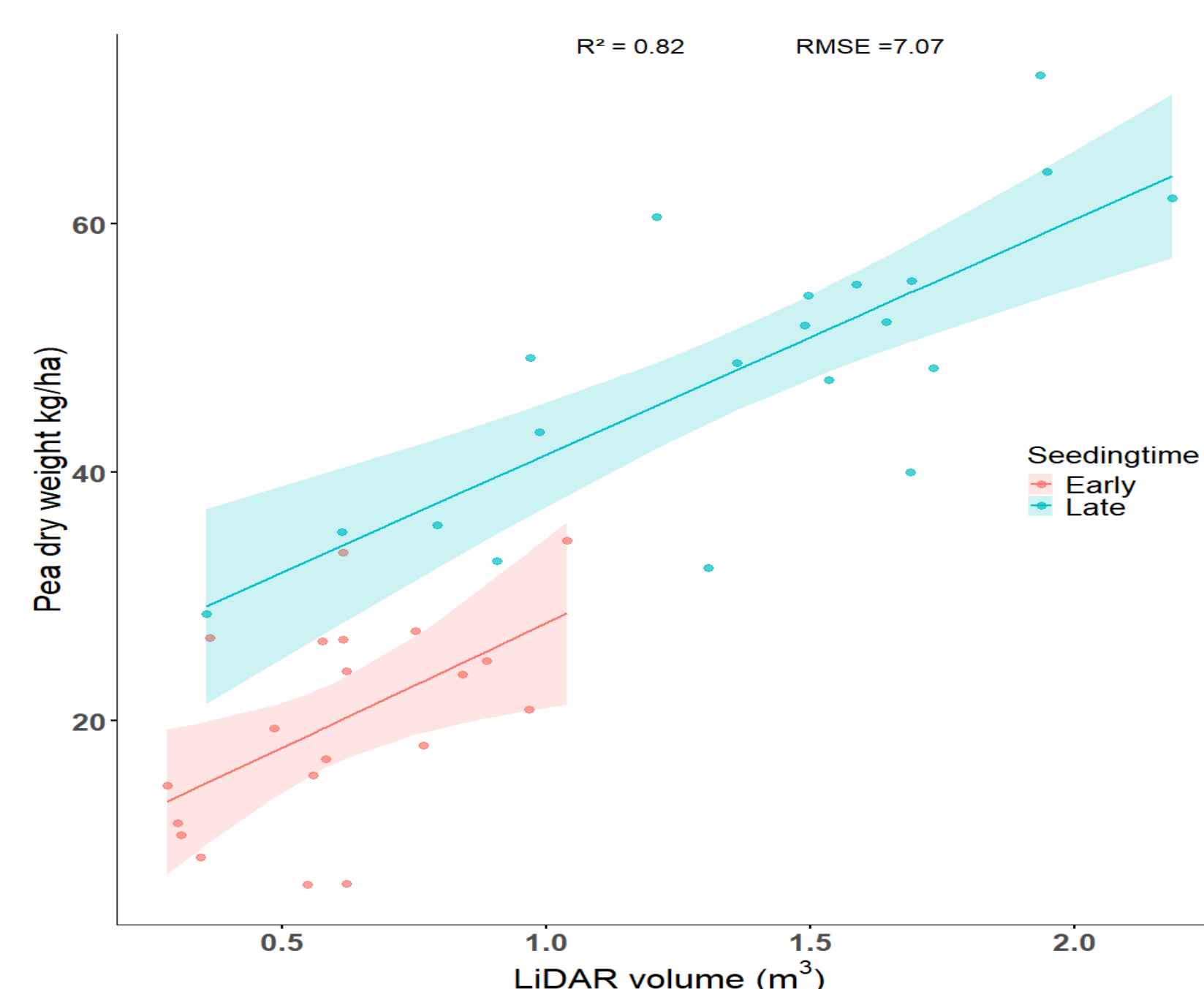


Figure 2: Relationship between the LiDAR point cloud-derived volume and the aboveground dry biomass

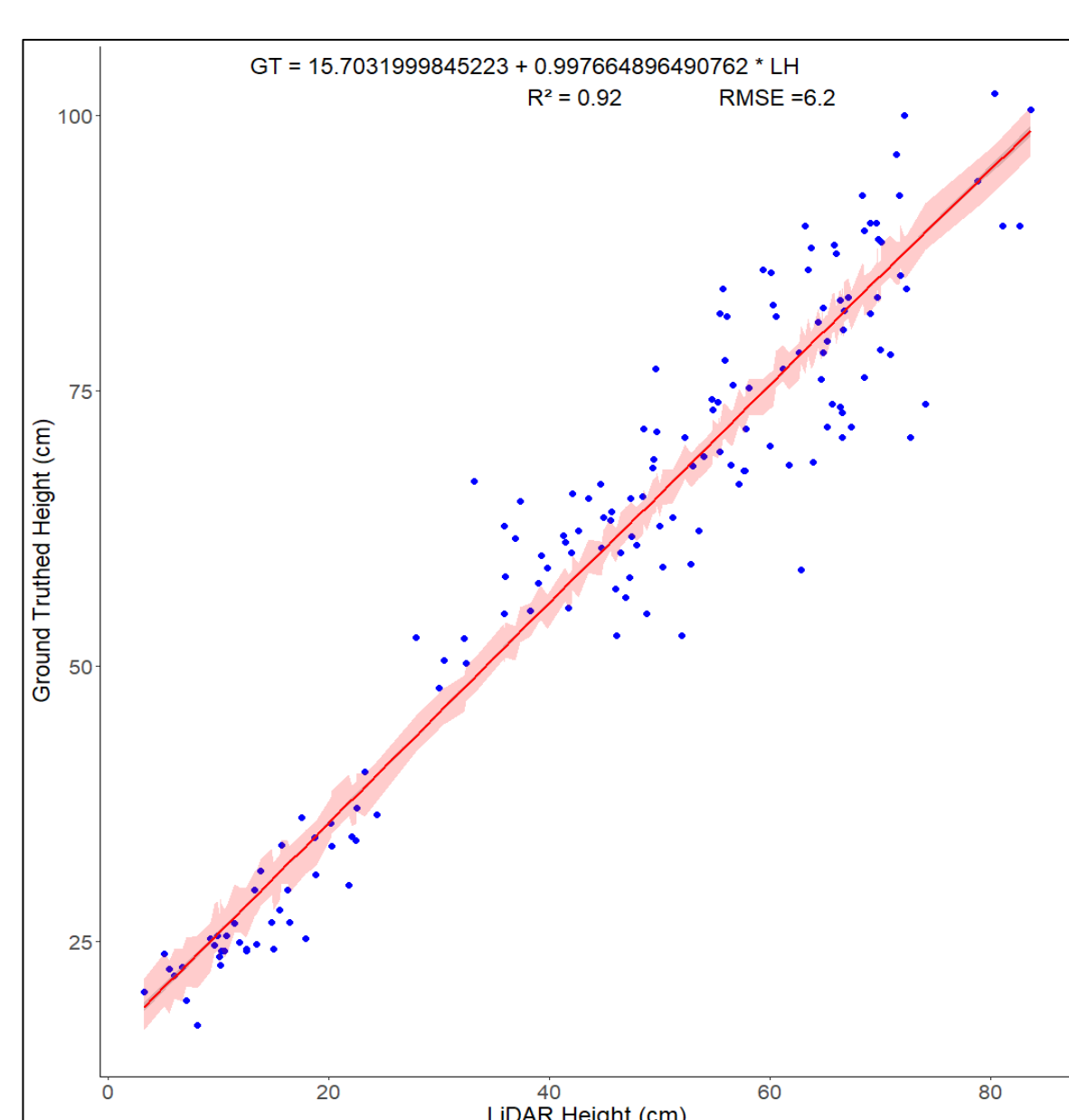


Figure 3: Relationship between the LiDAR point cloud-derived height and the ground truth data.

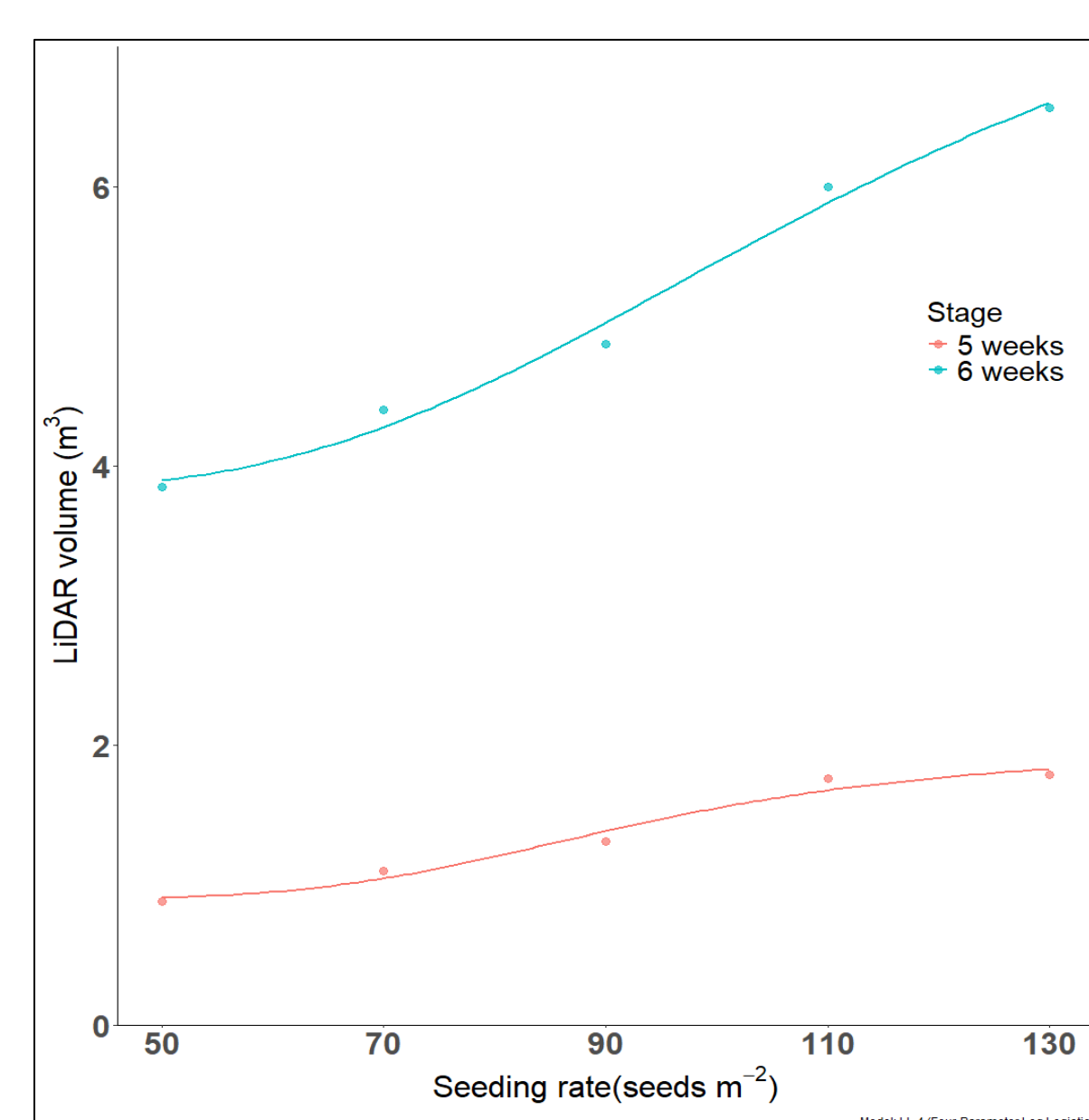


Figure 4: Response of LiDAR volume to increasing seeding rate

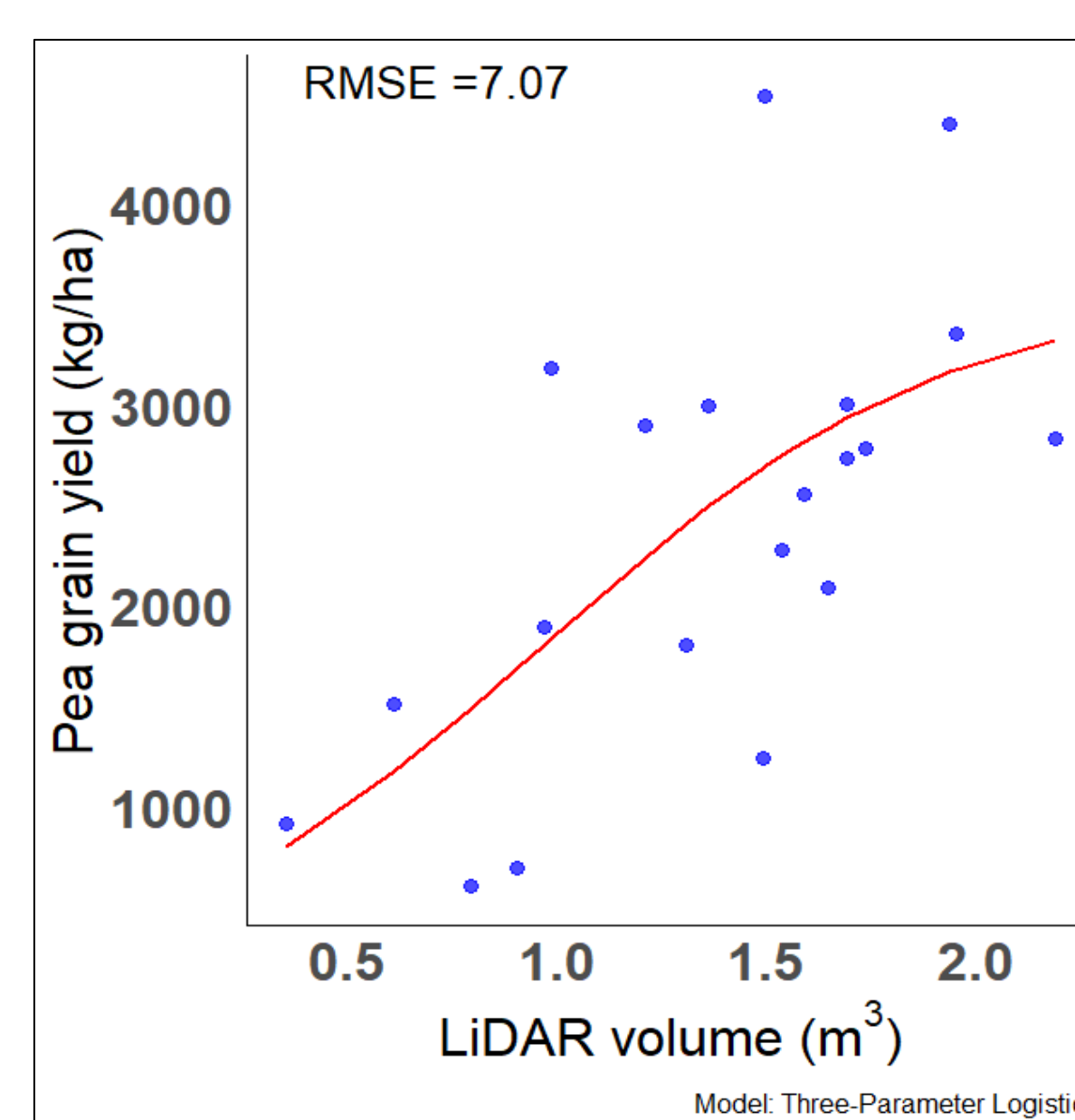
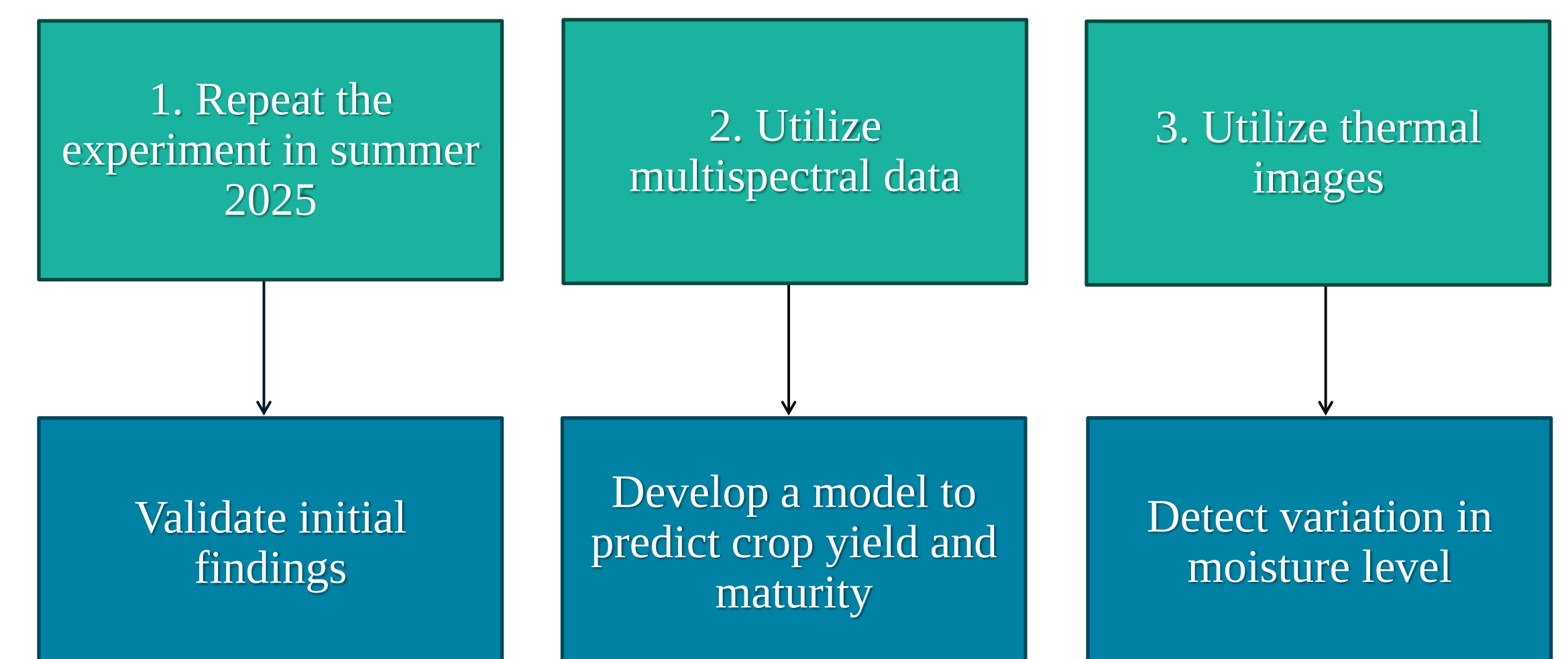


Figure 5: Relationship between LiDAR point cloud-derived volume and grain yield at V<sub>7</sub>-V<sub>8</sub> stage

## Conclusion

- This study demonstrates the effectiveness of LiDAR technology in evaluating best management practices (BMPs) for field pea.
- Preliminary results indicate that LiDAR can effectively distinguish between different BMPs, offering data-driven recommendations for optimizing seeding rates and sowing times.
- These findings contribute to precision agriculture by enhancing crop yield and sustainability.

## Future Work



## Acknowledgement



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